

Potential-dependent wettability of nickel in concentrated KOH: constant-potential molecular dynamics with a validated electrolyte force field

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Draft status. Sections 1–3 are drafted. Section 4 (force-field development and validation) reports *completed* results. Section 5 (the contact-angle campaign) is a structural placeholder marked **[PENDING]**; those simulations are queued behind the validation reported here. **No number in this manuscript is unsupported by a file in the project repository.** This is Paper 1 of two; the companion paper propagates the wettability reported here into a bubble-coverage closure for cell-scale electrolyser CFD.

Abstract

[PENDING] To be written once the contact-angle campaign completes. Structure fixed in advance: (i) molecular simulations of electrolytic bubbles are performed without electrode potential control, which their own authors identify as a limitation; (ii) we apply constant-potential molecular dynamics to hydroxide-rich electrolyte at a nickel electrode, and first establish a force field that reproduces bulk KOH properties; (iii) we show that reproducing the electrolyte density requires both a correctly sized hydroxide ion *and* scaled ionic charges, neither being sufficient alone; (iv) the resulting potential dependence of the nickel–electrolyte contact angle.

Keywords: constant-potential molecular dynamics; electrowetting; alkaline water electrolysis; hydroxide force field; electronic continuum correction; nickel electrode

1 Introduction

1.1 Bubble wettability is simulated without the potential that drives it

Gas bubbles nucleating at electrodes control much of the efficiency of water electrolysis, and the wettability of the electrode surface governs how large those bubbles grow before they detach. Molecular dynamics is the natural tool for interrogating that wettability, and it has been applied to electrolytic nanobubbles: on substrates of differing surface chemistry and curvature, and on surfaces engineered with patterned wettability.

These studies share a structural limitation, and it is one their authors state. Gas evolution is imposed as a prescribed conversion of liquid molecules into gas molecules at a fixed rate, which plays the role of an electrochemical reaction rate but, in the authors’ own words, “*can not be really assumed to mimic the electrochemical potential conditions*”. The electrode in these simulations is uncharged. Wettability is instead tuned through the solid–liquid interaction parameter, which reproduces a target contact angle but carries no electrochemical meaning: the parameter is adjusted until the desired angle appears, after which it plays no further role.

Separately, the methodology for imposing an electrode potential in classical molecular dynamics is mature and has been reviewed: constant-potential schemes solve for the induced charges on electrode atoms at every timestep so that each atom sits at its imposed potential.

These two literatures have not been joined. Potential control is available and catalogued; bubble wettability is simulated without it. The present work applies the former to the latter for the electrolyte and electrode of practical interest—concentrated KOH on nickel.

1.2 Why this cannot begin with the contact angle

The obvious approach is to build the interface and measure the angle. We report here why that approach fails, and what has to be established first.

An electrode–electrolyte contact angle is a property of the electrolyte as much as of the surface. It depends on the liquid–vapour surface tension, on the ion distribution near the interface, and on the density and structure of the solvent. A force field that misrepresents concentrated KOH in bulk cannot be trusted to represent it against a charged wall, where ion–surface interactions and the double-layer structure are precisely the quantities of interest.

We therefore treat bulk validation as a gate on the interface work, with criteria fixed before any result was inspected. Section 4 reports the outcome: an obvious and superficially reasonable hydroxide model **fails that gate**, in a way that is diagnostic rather than merely disqualifying, and the repair required is not the one that the failure first suggests.

1.3 Contributions

1. A demonstration that representing hydroxide as a single Lennard-Jones site carrying unit negative charge with SPC/E oxygen parameters—a natural and documented choice—produces an electrolyte whose apparent molar volume is *negative*, i.e. a model in which dissolving KOH in water causes it to contract.
2. Quantitative separation of the two candidate causes: we show that correcting the hydroxide *size* to its published value is necessary but **not sufficient**, leaving a 6–8 % density error, and that scaling the ionic charges is what closes the gap.
3. **[PENDING]** The potential dependence of the nickel–KOH contact angle from constant-potential molecular dynamics, with box-size convergence and replica statistics.

2 Theory

2.1 Constant-potential electrodes

Electrode atoms carry Gaussian charges that are not fixed but solved at every timestep so that each electrode atom sits at its imposed potential. With electrode charge vector $\mathbf{q}$ and elastance matrix $\mathbf{A}$,

$$\mathbf{A} \mathbf{q} = \mathbf{b}(\Psi, \{\mathbf{r}_{\text{electrolyte}}\}), \quad \sum_i q_i = 0, \quad \Delta\Psi = \Psi_{\text{anode}} - \Psi_{\text{cathode}} \quad (1)$$

The charge-neutrality constraint is enforced exactly. This is the step absent from studies that impose gas evolution at a prescribed rate on an uncharged wall.

2.2 Expected electrowetting response

The thermodynamic expectation follows from the Lippmann relation. Charging the electrode lowers the solid–liquid interfacial free energy, so

$$\cos \theta(\Delta\Psi) = \cos \theta_{\text{pzc}} + \frac{C_{\text{dl}} (\Delta\Psi - \Psi_{\text{pzc}})^2}{2\gamma_{\text{lv}}} \quad (2)$$

For a double-layer capacitance of 0.1–0.3 F/m² and $\gamma_{lv} \approx 0.072$ N/m, a 0.5 V excursion predicts a contact-angle shift of roughly 10°–31°. Equation (2) saturates unphysically at larger excursions; contact-angle saturation is a documented electrowetting phenomenon, so the quadratic is an upper bound and its breakdown would be a result rather than a failure.

2.3 Apparent molar volume as the diagnostic quantity

Density alone is a poor diagnostic for an electrolyte force field because it conflates the solvent model with the solute. The discriminating quantity is the apparent molar volume of the salt,

$$\phi_V = \frac{V_{\text{solution}} - n_w V_w^\circ}{n_{\text{KOH}}} \quad (3)$$

which isolates the volume the dissolved ions actually occupy, including electrostriction of their hydration shells. Section 4 shows that a model can miss the density by a modest-looking percentage while getting ϕ_V qualitatively wrong—indeed, of the wrong sign.

3 Methods

3.1 Code, electrostatics and potential control

LAMMPS (22 Jul 2025 stable), built from source with the ELECTRODE package, MPI and FFTW3. Long-range electrostatics use `pppm/electrode` with slab correction for the 2D-periodic geometry; `fix electrode/conp` solves Eq. (1).

The implementation was verified against the reference case distributed with the package, reproducing the published induced electrode charges **to all printed digits at every step of the potential ramp**. In the production geometry the constraint $\sum_i q_i = 0$ holds exactly at every timestep, with induced charges equal and opposite on the two slabs and growing in magnitude as the double layer forms.

3.2 Systems

Bulk validation systems contain 1500 SPC/E water molecules with K⁺ and OH[−] at 20 and 30 wt% KOH (120 and 206 ion pairs respectively), at 298 and 333 K, with three independent seeds per state point.

Interface systems place five-layer Ni(111) slabs ($a = 3.524$ Å) either side of an aqueous KOH film, with a cylindrical hydrogen bubble on the cathode where a contact angle is required. Lateral box widths of 8, 12 and 16 nm (19k, 30k and 41k atoms) are used for box-size convergence.

3.3 Initialisation

Lattice initial configurations are relaxed by a displacement-capped `nve/limit` push under a strong Langevin thermostat, **not** by energy minimisation. LAMMPS ignores SHAKE constraints during minimisation, so a minimised configuration is not consistent with the constrained dynamics that follow; conjugate-gradient minimisation with particle–mesh electrostatics on a lattice start was also found to be ill-conditioned, achieving 21 iterations in two minutes of wall time.

3.4 Contact-angle protocol

The bubble is cylindrical (quasi-2D) and periodic along its axis. The interface is located as the isochore of the time-averaged density field at $\rho = \frac{1}{2}(\rho_l + \rho_g)$, and the angle is measured *through the liquid*.

No line-tension extrapolation is performed, and none is possible in this geometry. A periodic cylindrical cap has straight contact lines with zero in-plane geodesic curvature, so the

modified Young relation $\cos\theta(R) = \cos\theta_\infty - \tau/(\gamma_{lv}R)$ —which is derived for a *circular* contact line—does not apply to it. Cylindrical geometry is used in the literature precisely because it *suppresses* the line-tension contribution. Fitting a $1/R$ trend to cylindrical caps of differing size would mistake finite-size and periodic-image artefacts for line tension.

Size dependence is instead addressed by box-size convergence across the three lateral widths above, with invariance required within replica scatter. Each state point uses at least three independent seeds, and reported uncertainty is the *spread across replicas*: block averages taken from a single trajectory are not independent realisations.

The control variable is the electrode potential referenced to the potential of zero charge, determined in a separate calculation. The raw inter-slab potential difference is a cell voltage, not an electrode potential.

3.5 The validation gate

Criteria were fixed before any result was inspected: density within 3 %, $D(\text{H}_2\text{O})$ within 30 %, $D(\text{K}^+)$ within 40 %. Diffusivities are obtained from the Einstein relation over the linear region of the mean-squared displacement with the Yeh–Hummer finite-size correction, $D_\infty = D_{\text{PBC}} + k_B T \xi / (6\pi\eta L)$ with $\xi = 2.837297$; measured corrections are 10–17 %, largest at high concentration where the box is smallest and the viscosity highest.

Hydroxide diffusivity is reported but deliberately not gated. No classical fixed-charge model can reproduce structural (Grotthuss) transport, so $D(\text{OH}^-)$ is expected to be too slow. Gating on it would be a category error; omitting the statement would be worse.

4 Results: force-field development

4.1 An obvious hydroxide model fails, and fails informatively

Hydroxide was first represented as a single Lennard-Jones site carrying unit negative charge with parameters identical to SPC/E oxygen ($\sigma = 3.166 \text{ \AA}$, $\varepsilon = 0.1553 \text{ kcal/mol}$), following documented practice for hydroxide in SPC/E-based simulations. Potassium used Joung–Cheatham parameters. This model failed the density gate at all four state points (Table 1, Fig. 1).

Table 1: Bulk KOH(aq) validation of the initially adopted model. Three independent seeds per state point; uncertainty is the replica spread. Pre-registered tolerance 3 %.

System	ρ_{sim} (g/cm ³)	ρ_{exp}	Error	$D_{\text{H}_2\text{O}}$ (m ² /s)	Verdict
20 wt%, 298 K	1.2672 ± 0.0020	1.185	+6.94 %	9.74×10^{-10}	FAIL
20 wt%, 333 K	1.2367 ± 0.0021	1.166	+6.06 %	2.00×10^{-9}	FAIL
30 wt%, 298 K	1.4081 ± 0.0048	1.286	+9.50 %	3.90×10^{-10}	FAIL
30 wt%, 333 K	1.3817 ± 0.0008	1.265	+9.23 %	1.00×10^{-9}	FAIL

The failure is **systematic rather than statistical**: replica scatter is 0.002–0.005 g/cm³ against a discrepancy of 0.08–0.12 g/cm³, a factor of 20–60.

4.2 The apparent molar volume states the failure correctly

The natural reading of Table 1 is that the error grows with concentration—6.9 % at 20 wt% to 9.5 % at 30 wt%—and therefore implicates the ions. **That inference is unsafe.** Converting to apparent molar volume via Eq. (3) shows the deficit is approximately *constant per mole of salt*:

The growth of the percentage error with concentration is therefore largely arithmetic: more ions multiplying a roughly constant per-ion defect. Stated properly, **the model says that dissolving KOH in water causes the liquid to contract**. The volume deficit of $\approx 13 \text{ cm}^3/\text{mol}$ corresponds to about $22 \text{ \AA}^3$ per formula unit, and to a spurious internal pressure of order 2.5 kbar.

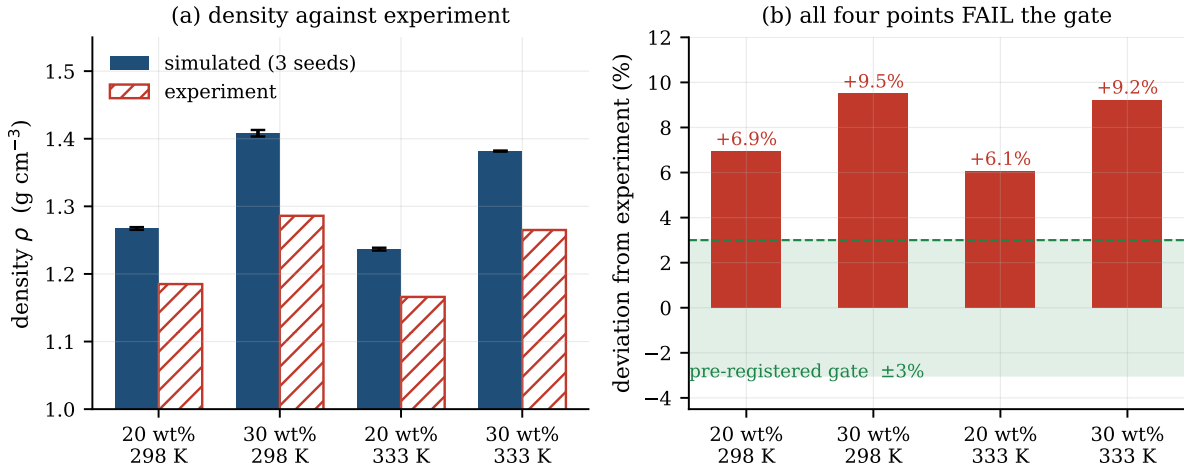


Figure 1: Bulk KOH density of the initially adopted model against experiment. (a) Simulated density, three seeds, error bars are replica spread. (b) Deviation from experiment; shaded band is the pre-registered $\pm 3\%$ tolerance.

Table 2: Apparent molar volume of KOH. The model assigns essentially zero, and at 20 wt% *negative*, volume to the dissolved salt.

	ϕ_V simulated (cm^3/mol)	ϕ_V experiment	Deficit
20 wt%, 298 K	-3.5	+11.5	15.0
30 wt%, 298 K	+1.6	+14.0	12.4

The physical origin is straightforward once stated this way. SPC/E’s oxygen parameters are calibrated for a site carrying -0.8476 that is *shielded* by two hydrogens at $1 \text{ \AA}$. Assigning that same repulsive core a bare, unshielded -1.0 strengthens the ion–water attraction by roughly 18% with no compensating increase in core size, and the first hydration shell collapses inward.

4.3 Size is necessary but not sufficient

Two repairs are available: enlarge the hydroxide, or reduce the effective ionic charge. They are not distinguishable from density at two concentrations alone, and we therefore tested them separately (Fig. 2).

A scan of the hydroxide Lennard-Jones diameter at fixed well depth (Fig. 2a) crosses the experimental density at $\sigma = 3.67 \text{ \AA}$, and $\sigma = 3.70 \text{ \AA}$ reproduces the 30 wt% density to -0.57% . **On the density criterion alone, size tuning appears to solve the problem.** The crossing also lies close to the independently published value of $3.81 \text{ \AA}$ derived from solvation free energies and activities, which is reassuring as to the diagnosis.

It does not solve the problem. Adopting the published diameter and retaining formal ± 1 charges still overpredicts the density by 6.0% at 20 wt% and 7.8% at 30 wt% (Fig. 2b). Scaling the ionic charges to ± 0.8 , in the spirit of the electronic continuum correction, brings both concentrations to within 1.5%.

4.4 Why the size-only repair was rejected

The size-tuned model passes the density gate. It was not adopted.

Tuning a single Lennard-Jones diameter to reproduce one target density absorbs a charge-related error into a size parameter. The resulting hydroxide reproduces the density by construction while having a hydration-shell radius, solvation free energy and coordination number that are wrong in compensating directions. Such a model is fit for reproducing the quantity it was fitted

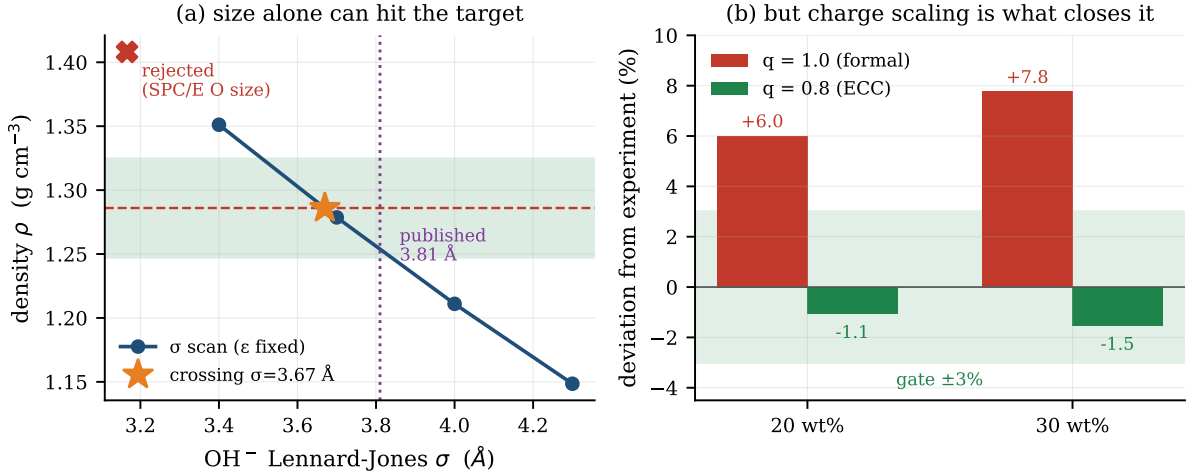


Figure 2: Force-field development. (a) Scanning the hydroxide Lennard-Jones diameter at fixed well depth crosses the experimental density (dashed, with the $\pm 3\%$ gate shaded) at $\sigma \approx 3.67 \text{ \AA}$, close to the independently published $3.81 \text{ \AA}$ (dotted). The rejected model is marked. (b) At the published diameter, formal ± 1 ionic charges still overpredict density by 6–8%; scaling to ± 0.8 brings both concentrations inside the gate. Size is necessary but not sufficient.

to and little else—and in particular it is not fit for an interface calculation, where ion size governs surface propensity and the distribution of ions relative to the wall is the quantity of interest.

This is the substantive reason for reporting a validation gate at all. A gate that is passed by adjusting a parameter until it is passed has not tested anything. The distinction between the two repairs is invisible in the density and visible in the physics.

The adopted parameters are given in Table 3.

Table 3: Adopted force field. Water is SPC/E with charges *unscaled*, SPC/E being already an effective-charge model; ion charges are scaled by 0.8. Lennard-Jones parameters are combined geometrically.

Site	ε (kcal/mol)	σ ($\text{\AA}$)	q (e)
O (water, SPC/E)	0.15530	3.16600	-0.8476
H (water, SPC/E)	0	0	$+0.4238$
K^+	0.21510	2.83000	$+0.8$
OH^- (single site)	0.01195	3.81000	-0.8

We note that the optimal hydrogen charge for hydroxide in this class of model has been found to be exactly zero, so the single-site representation is not an approximation adopted for convenience: it is the form the optimisation selects. **Our error was in the parameter values, not in the topology.**

4.5 Re-validation

[PENDING] Full 12-case re-validation with the adopted force field, including transport properties and the hydroxide–water radial distribution function. The first completed cases give 1.1731 g/cm^3 at 20 wt% (-1.0%) and 1.2688 g/cm^3 at 30 wt% (-1.3%), both inside the gate; the remaining cases and the transport stage are running at the time of writing.

5 Results: potential-dependent wettability

[PENDING] Double-layer structure and differential capacitance against applied potential.

[PENDING] Contact angle $\theta(\Delta\Psi)$ referenced to the potential of zero charge, with replica

statistics.

[PENDING] Box-size convergence across 8, 12 and 16 nm lateral widths.

[PENDING] Comparison with the Lippmann prediction of Eq. (2), including any evidence of contact-angle saturation.

[PENDING] Surface tension and dissolved-gas diffusivity for propagation to the companion study.

6 Limitations

The model is chemically inert. SPC/E water with classical ions on a clean Ni(111) surface cannot represent nickel oxidation, hydroxylation, specific adsorption or bond formation. Constant-potential dynamics makes the metal *charge* responsive; it does not make the interface *chemically* reactive. What is reported is the electrostatic, double-layer contribution to potential-dependent wettability on an idealised surface—not a prediction of the wettability of an operating electrode.

Hydroxide transport is structurally limited. A classical fixed-charge OH^- cannot carry Grotthuss transport, so its diffusivity and the ionic conductivity are reported but excluded from all validation criteria, permanently and by design.

Scaled charges are an effective-medium device. The electronic continuum correction represents electronic polarisability by reducing the ionic charge; it does not restore the polarisation response itself, and quantities sensitive to that response should not be taken from this model without independent checking.

An equilibrium angle is not a dynamic one. The reported quantity is the equilibrium contact angle and its potential-induced shift. Advancing and receding angles, and the hysteresis between them, are governed by roughness and chemical heterogeneity that a flat single crystal does not contain.

Data availability

All inputs, analysis scripts, force-field files and the rejected-model runs are available in the accompanying repository. **The runs of the rejected force field are retained deliberately**, being the evidence base for Section 4. [PENDING] Repository DOI to be minted on submission.

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